# Principles of Quantum Mechanics

Part II

Recorded from the lecturer in 2026-2027.

## 0. Introduction

∗ Recall features from IB QM (Supplementary Notes 1)

- Wave-particle duality:

  Waves behaving like particles, e.g. light quanta, & vice versa (particles behaving like waves).

- Wavefunction $\psi(x)$ for a particle — prob. density $|\psi(x)|^2$.

  (Prob. intrinsic to theory.)

- Observables became operators on wavefns:

$$\hat{x} \to x, \qquad \hat{p} \to -i\hbar \frac{\partial}{\partial x}.$$

  & lack of commutativity, e.g.

$$[\hat{x}, \hat{p}] = i\hbar$$

  leads to Uncertainty Relation.

- Schrödinger eqn with Hamiltonian $H$ specifies dynamics & eigenvalues of $H$ are allowed energy levels of system.

- All of this explains H atom structure & resulting line spectra.

∗ Aims of this course:

- Reformulate QM in a more powerful, flexible, & useful form — the Dirac formalism (go beyond wavefns!).

Previous description appears as particular realisation & also leads to alternatives, e.g.

$$\tilde{\psi}(p) \qquad \text{with} \qquad \hat{x} \to +i\hbar \frac{\partial}{\partial p} \quad \& \quad \hat{p} \to p.$$

This more general approach allows

- Simpler analysis of known problems, e.g. oscillator

but also lots of new things

- Spin of particles
- Symmetries (translations, rotations, for example) & conservation laws.
- Identical particles (bosons & fermions)
- Framework for quantising more general systems, e.g. EM field
- Perturbation theory: practical tool for computations — we will not look at too many detailed applications, but we will try to keep track of what all this is for — physical predictions

# 1. Dirac Formalism

## 1.1 Concepts & Definitions

A quantum system is described at each instant by a state $|\psi\rangle$ which belongs to a complex vector space $V$; any non-zero element of this space $V$ is a possible state.

Thus

$$|\psi\rangle \;\&\; |\phi\rangle \in V \quad \Rightarrow \quad \alpha|\psi\rangle + \beta|\phi\rangle \in V \qquad \text{for any } \alpha, \beta \in \mathbb{C}.$$

Physically this is a superposition principle leading to wave-like behaviour, but these states are not wavefns. There are also dual or conjugate states $\langle\phi|$ which belong to the dual space $V^\dagger$.

By definition, states & duals can be combined or paired to give a complex number

$$\underset{\text{"bra"}}{\langle\phi|}\,, \quad \underset{\text{"ket"}}{|\psi\rangle} \longmapsto \underset{\text{"bra(c)ket"}}{\langle\phi|\psi\rangle}\,, \qquad V^\dagger \times V \to \mathbb{C},$$

with

$$\langle\phi|\big(\alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle\big) = \alpha_1\langle\phi|\psi_1\rangle + \alpha_2\langle\phi|\psi_2\rangle,$$
$$\big(\beta_1\langle\phi_1| + \beta_2\langle\phi_2|\big)|\psi\rangle = \beta_1\langle\phi_1|\psi\rangle + \beta_2\langle\phi_2|\psi\rangle.$$

The spaces $V$ & $V^\dagger$ come with an inner product, which can be described as a one-to-one correspondence between the spaces

$$V \longleftrightarrow V^\dagger,$$

with

$$|\psi\rangle \longleftrightarrow \langle\psi| = \big(|\psi\rangle\big)^\dagger \qquad \text{(use same symbol for corresponding states)}.$$

The inner product is

$$|\phi\rangle,\ |\psi\rangle \longmapsto \langle\phi|\psi\rangle = \big(|\phi\rangle\big)^\dagger|\psi\rangle, \qquad V \times V \to \mathbb{C},$$

& it is assumed to obey

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^* \qquad \text{hermitian},$$
$$\big\|\,|\psi\rangle\,\big\|^2 = \langle\psi|\psi\rangle \qquad \text{real},$$
$$\left.\begin{array}{l}\langle\psi|\psi\rangle \geq 0,\\ \langle\psi|\psi\rangle = 0 \quad \text{iff} \quad |\psi\rangle = 0\end{array}\right\} \qquad \text{positive definite},$$

where now $\big\|\,|\psi\rangle\,\big\|$ is called the norm.

Note: Knowing $\langle\phi|\psi\rangle$ for all $\langle\phi|$ determines $|\psi\rangle$ (& vice versa).

[Check:

$$\langle\phi|\psi_1\rangle = \langle\phi|\psi_2\rangle \quad \Rightarrow \quad \langle\phi|\chi\rangle = 0, \qquad \text{where } |\chi\rangle = |\psi_1\rangle - |\psi_2\rangle.$$

If this holds $\forall\,\langle\phi|$, set $\langle\phi| = \langle\chi|$ & get

$$\langle\chi|\chi\rangle = \big\|\,|\chi\rangle\,\big\|^2 = 0 \quad \Rightarrow \quad |\chi\rangle = 0, \qquad \text{as claimed.}$$

]

The physical content of any state is unchanged by

$$|\psi\rangle \to \alpha|\psi\rangle \qquad \text{for any } \alpha \in \mathbb{C}, \quad \alpha \neq 0.$$

We will usually normalise to make $\big\|\,|\psi\rangle\,\big\| = 1$, but can still change

$$|\psi\rangle \to e^{i\theta}|\psi\rangle.$$

Relative phases can be important in expressions like $\alpha|\psi\rangle + \beta|\phi\rangle$.

The space $V$ is complete: we assume that appropriate sequences & series converge. A complete inner product space is called a Hilbert space.

An operator $Q$ is a linear map on states

$$|\psi\rangle \longmapsto Q|\psi\rangle, \qquad V \to V,$$

with

$$Q\big(\alpha|\psi\rangle + \beta|\phi\rangle\big) = \alpha Q|\psi\rangle + \beta Q|\phi\rangle.$$

The same operator can be thought of as acting to the left on dual states,

$$\langle\phi| \longmapsto \langle\phi|Q, \qquad V^\dagger \to V^\dagger,$$

where the right-hand side is defined by

$$\big(\langle\phi|Q\big)|\psi\rangle = \langle\phi|\big(Q|\psi\rangle\big) = \langle\phi|Q|\psi\rangle.$$

**End of Lecture 1**

Last time: abstract notation for states

$$|\psi\rangle \in V, \qquad \langle\phi| \in V^\dagger, \qquad \& \text{ map } V \longleftrightarrow V^\dagger,$$

& operators acting, e.g. $|\psi\rangle \longmapsto Q|\psi\rangle$ (§ 1.1).

Note: States & operators $|\psi\rangle$, $\langle\phi|$, $Q$ can be dealt with by rules that mirror those for complex vectors $\underline{u}$ (column), $\underline{v}^\dagger$ (row), & a matrix $A$; e.g. compare

$$\langle\phi|Q|\psi\rangle \qquad \text{with} \qquad \underline{v}^\dagger A \underline{u}.$$

For any operator $Q$, say $|\psi\rangle \neq 0$ is an eigenstate with eigenvalue $\lambda$ if

$$Q|\psi\rangle = \lambda|\psi\rangle;$$

equivalently,

$$\langle\psi|Q^\dagger = \lambda^*\langle\psi|.$$

For general $Q$, $\lambda$ may be complex.

For any $Q$, the hermitian conjugate or adjoint $Q^\dagger$ is defined by

$$\langle\phi|Q^\dagger = \big(Q|\phi\rangle\big)^\dagger \qquad \text{for any } |\phi\rangle.$$

Equivalently,

$$\begin{aligned}\langle\phi|Q^\dagger|\psi\rangle &= \big(Q|\phi\rangle\big)^\dagger|\psi\rangle \\ &= \langle\psi|Q|\phi\rangle^* \qquad \text{for all } |\phi\rangle,\ |\psi\rangle.\end{aligned}$$

Simple consequences:

$$\begin{aligned}(\alpha A + \beta B)^\dagger &= \alpha^* A^\dagger + \beta^* B^\dagger, \\ (AB)^\dagger &= B^\dagger A^\dagger,\end{aligned}$$

for any operators $A, B$ & $\alpha, \beta \in \mathbb{C}$, &

$$\big(Q^\dagger\big)^\dagger = Q.$$

Observables

An operator $Q$ is hermitian or self-adjoint if

$$Q^\dagger = Q;$$

such operators are called observables because they correspond to physical quantities that can be measured, e.g. position, momentum, energy.

To define a quantum system, we must specify the Hamiltonian $H$ whose eigenvalues are the allowed energy levels of the system:

$$H|\chi\rangle = E|\chi\rangle \qquad \text{energy eigenvalue problem.}$$

Key results for any hermitian $Q$:

(i) All eigenvalues are real.

(ii) Eigenstates with distinct eigenvalues are orthogonal.

(iii) Any state can be expanded in terms of (written as a linear combination of) eigenstates.

We prove (i) & (ii) & assume (iii).

(i) Suppose

$$Q|\psi\rangle = \lambda|\psi\rangle.$$

Then

$$\langle\psi|Q^\dagger = \lambda^*\langle\psi| = \langle\psi|Q \qquad (Q^\dagger = Q).$$

Hence

$$\langle\psi|Q|\psi\rangle = \lambda\langle\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle.$$

But

$$\big\| \, |\psi\rangle \, \big\|^2 = \langle\psi|\psi\rangle \neq 0 \qquad (|\psi\rangle \neq 0),$$

so $\lambda^* = \lambda$.

(ii) Let $|n\rangle$ be eigenstates of $Q$ with eigenvalues $\lambda = q_n$ (real), where $n$ is some discrete label, possibly ranging over infinitely many values.

$$Q|n\rangle = q_n|n\rangle, \qquad Q|m\rangle = q_m|m\rangle,$$

or

$$\langle m|Q = q_m\langle m| \qquad (q_m^* = q_m).$$

Hence

$$\langle m|Q|n\rangle = q_n\langle m|n\rangle = q_m\langle m|n\rangle.$$

But $q_m \neq q_n$, so $\langle m|n\rangle = 0$.

## 1.2 The Harmonic Oscillator

A one-dimensional harmonic oscillator of mass $m$ & frequency $\omega$ is a system with position $\hat{x}$, momentum $\hat{p}$ & Hamiltonian

$$H = \frac{1}{2m}\hat{p}^{\,2} + \frac{1}{2}m\omega^2\hat{x}^{\,2}.$$

Seek eigenstates & eigenvalues of $H$.

(a) Annihilation, creation & number operators

Define

$$\begin{cases} a = \left(\frac{m\omega}{2\hbar}\right)^{1/2}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \\ a^\dagger = \left(\frac{m\omega}{2\hbar}\right)^{1/2}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right). \end{cases}$$

These are dimensionless hermitian conjugates, since $\hat{x}^\dagger = \hat{x}$, $\hat{p}^\dagger = \hat{p}$. Equivalently,

$$\begin{cases} \hat{x} = \left(\frac{\hbar}{2m\omega}\right)^{1/2}(a + a^\dagger), \\ \hat{p} = \left(\frac{\hbar m\omega}{2}\right)^{1/2} i(a^\dagger - a). \end{cases}$$

Easy to check:

$$[\hat{x}, \hat{p}] = i\hbar \quad \Longleftrightarrow \quad [a, a^\dagger] = 1.$$

In fact,

$$\begin{aligned} aa^\dagger &= \frac{m\omega}{2\hbar}\hat{x}^{\,2} + \frac{1}{2m\omega\hbar}\hat{p}^{\,2} - \frac{i}{2\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x}) \\ &= \frac{1}{\hbar\omega}H + \frac{1}{2}, \end{aligned}$$

but

$$a^\dagger a = \frac{1}{\hbar\omega}H - \frac{1}{2}.$$

This confirms the commutation relation, but we also find

$$H = \hbar\omega\left(N + \frac{1}{2}\right),$$

with $N = a^\dagger a$ the number operator.

$N^\dagger = N$, & finding eigenvalues & eigenstates of $N$ is equivalent to finding those for $H$. Let $|\lambda\rangle$ be any eigenstate of $N$ with eigenvalue $\lambda$:

$$N|\lambda\rangle = \lambda|\lambda\rangle \quad \Longleftrightarrow \quad H|\lambda\rangle = E|\lambda\rangle, \qquad E = \hbar\omega\left(\lambda + \frac{1}{2}\right).$$

Then, assuming $\big\|\,|\lambda\rangle\,\big\|^2 = 1$ (normalised),

$$\begin{aligned} \lambda = \langle\lambda|N|\lambda\rangle &= \langle\lambda|a^\dagger a|\lambda\rangle = \big(a|\lambda\rangle\big)^\dagger\big(a|\lambda\rangle\big) \\ &= \big\|a|\lambda\rangle\big\|^2. \end{aligned}$$

Thus $\lambda \geq 0$ (all eigenvalues are non-negative), & $\lambda = 0$ iff $a|\lambda\rangle = 0$.

Now consider

$$\begin{aligned} N\big(a^\dagger|\lambda\rangle\big) &= \big([N, a^\dagger] + a^\dagger N\big)|\lambda\rangle \\ &= \big(a^\dagger + a^\dagger\lambda\big)|\lambda\rangle \\ &= (\lambda + 1)\big(a^\dagger|\lambda\rangle\big). \end{aligned}$$

Similarly,

$$\begin{aligned} N\big(a|\lambda\rangle\big) &= \big([N, a] + aN\big)|\lambda\rangle \\ &= (-a + a\lambda)|\lambda\rangle \\ &= (\lambda - 1)a|\lambda\rangle. \end{aligned}$$

Here we have used the results for commutators

$$[N, a^\dagger] = a^\dagger \qquad \big(= [a^\dagger a, a^\dagger] = a^\dagger [a, a^\dagger]\big),$$

&

$$[N, a] = -a.$$

New states will be eigenstates provided their norms are non-zero. Compute:

$$\big\|a|\lambda\rangle\big\|^2 = \lambda,$$

$$\begin{aligned}\big\|a^\dagger|\lambda\rangle\big\|^2 &= \langle\lambda|aa^\dagger|\lambda\rangle = \langle\lambda|a^\dagger a + 1|\lambda\rangle \\ &= \lambda + 1.\end{aligned}$$

Therefore $a^\dagger$ & $a$ are called <u>creation</u> (or raising) & <u>annihilation</u> (or lowering) operators, respectively. Since all eigenvalues are non-negative, any non-integer $\lambda$ would eventually lead to a negative eigenvalue under repeated lowering, which is impossible. Thus, $\lambda$ must be a non-negative integer. Repeated lowering then reaches the ground state $|0\rangle$, where $a|0\rangle = 0$. Since the raising operator $a^\dagger$ can act indefinitely, all non-negative integers are eigenvalues. Hence,

$$\boxed{\lambda = 0, 1, 2, \ldots}$$

**End of Lecture 2**

From calculations of norms above, we can choose normalised eigenstates $|n\rangle$ related by

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad a|n\rangle = \sqrt{n}\,|n-1\rangle.$$

Starting from any state $|n\rangle$ we can generate all others. In particular, starting from the ground state $|0\rangle$ characterised by

$$a|0\rangle = 0,$$

have normalised eigenstate

$$|n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n|0\rangle.$$

$a$, $a^\dagger$ also called <u>ladder operators</u>.

(b) <u>Importance of the oscillator</u>

- A smooth potential $V(x)$ with equilibrium point $x = x_0$ (so that $V'(x_0) = 0$) can be expanded

  $$V(x_0 + x) = V(x_0) + \frac{1}{2}V''(x_0)x^2 + O(x^3).$$

  So, if $x = x_0$ is <u>stable</u>, can set $V''(x_0) = m\omega^2$ & for displacements $x$ not too large, expect system to be well-approximated by oscillator.

  e.g. diatomic molecules — quantisation of vibrational energies important in understanding internal energy & heat capacities — macroscopic consequences.

- More complicated systems can be analysed in terms of <u>normal modes</u>: coherent motion in which all degrees of freedom oscillate with same common frequency $\omega$. General solution for <u>classical</u> oscillator is

  $$x(t) = Ae^{-i\omega t} + A^* e^{+i\omega t} \qquad A \text{ complex constant.}$$

Normal modes for a system with variables $x_N$ are solutions of form

$$x_N(t) = Au_N e^{-i\omega t} + A^* u_N^* e^{+i\omega t}$$

with $u_N$ complex vector specific to each normal frequency $\omega$.

e.g. crystal with # atom $\mathcal{N} \sim 10^{23}$ — forces approximately elastic & $N$ labels directions of motion for each atom, so $3 \times \mathcal{N}$ values.

or EM field — normal modes oscillations of electric & magnetic fields at each point in space

$$\underline{E}(\underline{x}, t) = A\underline{u}(\underline{x})e^{-i\omega t} + A^*\underline{u}(\underline{x})^* e^{+i\omega t}$$

for some complex vector field $\underline{u}(\underline{x})$. This is exact.

- QM: normal modes are independent oscillators — treat these as above. For each mode have

$$a(\omega) \quad \& \quad a(\omega)^\dagger.$$

Ground state $|0\rangle$ with

$$a(\omega)|0\rangle = 0.$$

Other states e.g. $a^\dagger(\omega)|0\rangle$ & find they have energy & momentum & behave like particles.

$$\left.\begin{matrix} a \text{ destroys} \\ a^\dagger \text{ creates} \end{matrix}\right\} \text{particles} \begin{cases} \text{phonons in crystal} \\ \text{photons in EM field} \end{cases}$$

### 1.3 Bases & Measurements

(a) Bases of Eigenstates

From key properties (i), (ii), (iii) in § 1.1 we know that for any observable $Q$ there is an orthonormal basis of eigenstates $\{|n\rangle\}$ for $V$

$$Q|n\rangle = q_n|n\rangle, \qquad \langle m|n\rangle = \delta_{mn}.$$

For any state,

$$|\psi\rangle = \sum_n \alpha_n |n\rangle, \qquad \text{where} \quad \alpha_n = \langle n|\psi\rangle.$$

Note

$$\big\| \, |\psi\rangle \, \big\|^2 = 1 \quad \Longleftrightarrow \quad \begin{aligned} \langle\psi|\psi\rangle &= \left(\sum_m \alpha_m^* \langle m|\right)\left(\sum_n \alpha_n |n\rangle\right) \\ &= \sum_n |\alpha_n|^2 = 1. \end{aligned}$$

Note that we can write

$$1 = \sum_n |n\rangle\langle n| \qquad \text{“completeness relation”}$$

since on any state

$$\begin{aligned} |\psi\rangle &= \left(\sum_n |n\rangle\langle n|\right)|\psi\rangle \\ &= \sum_n |n\rangle\big(\langle n|\psi\rangle\big), \end{aligned}$$

so holds for any orthonormal basis.

In above there may be several independent states with same eigenvalue $\lambda$. The eigenspace for a given eval is

$$V_\lambda = \big\{|\psi\rangle : Q|\psi\rangle = \lambda|\psi\rangle\big\}.$$

& note $V_\lambda$ has basis $\big\{|n\rangle : q_n = \lambda\big\}$. The degeneracy of $\lambda$ is the number of states in this basis, i.e. $\dim V_\lambda$, but note that $\lambda$ is also called non-degenerate if degeneracy is 1.

**End of Lecture 3**